

6

Economics of Power Generation

Load Curve: It is plot of load in kilowatts versus time (usually for a day or a year.) in the order in which they occur, i.e., chronologically. The consumer consumes the electrical energy when he needs. This means his load on the system will keep on changing with time. The curve drawn on a daily basis is called **daily load curve** or the curve drawn on an annual basis is called **Annual load curve**. The average load on the supply system will be less than the maximum demand.

Characteristics:

1. The load can be determined at any particular hour of the day.
2. The maximum and minimum values of the daily load can be found out from the load curve.
3. The area under the daily load curve gives the energy supplied in a day (month or year if considered monthly or yearly load curves).
4. The daily average load can be found out by calculating the area under the load curve (i.e. energy in kWh) and dividing it by the number of hours (i.e. 24 hours).
5. Information whether the power system is working efficiently or not.

Load Duration Curve: It is the plot of load in kilowatts versus time duration for which it occurs in the descending order of magnitude, irrespective of the time of occurrence

Load Factor: The ratio of average load to the maximum demand during a given period is known as load factor.

$$\text{Load factor} = \frac{\text{Average load}}{\text{Maximum demand}}$$

Average Load: The average of loads occurring on the power station in a given period is known as average load or average demand.

$$\text{Daily avg load} = \frac{\text{No. of units (kWh) generated in a day}}{24 \text{ hours}}$$

$$\text{Monthly avg load} = \frac{\text{No. of units generated in month}}{\text{No. of hours in a month}}$$

$$\text{Yearly avg load} = \frac{\text{No. of units generated in a year}}{8760 \text{ hours}}$$

Maximum Demand: It is the greatest demand of load on power station during a given period.

Demand Factor: It is the ratio of maximum demand on the power station to its connected load,

$$\text{Demand factor} = \frac{\text{Maximum Demand}}{\text{Connected Load}}$$

Connected Load: It is the sum of continuous ratings of all the equipments connected to the supply system.

Diversity Factor: The ratio of the sum of individual maximum demands to the simultaneous maximum demand on the power station is known as diversity factor.

Diversity factor

$$\text{Diversity factor} = \frac{\text{Sum of individual maximum demands}}{\text{Simultaneous Maximum demand on power station}}$$

- Diversity factor is higher, the energy is cheaper.
- Load factor, diversity factor higher the energy per unit is less.
- Diversity factor has direct effect on the fixed cost of unit generated.

Coincidence Factor: It is the reciprocal of diversity factor and is always less than '1'.

Plant capacity factor: It is defined as the ratio of average demand on the station to the maximum installed capacity.



Capacity factor =

$$\frac{\text{Actual energy produced}}{\text{Maximum energy that could have been produced}}$$

Plant capacity factor will be the ratio of the total energy generated over the year, to the plant capacity (including reserve capacity) multiplied by 8760 hours.

Reserve capacity = Plant capacity – Maximum demand.

Operation Factor (Service Factor): It is given by the ratio of number of hours the plant is in service to the total number of hours in a given period (usually a year)

$$\text{Operation factor} = \frac{\text{Service hours}}{\text{Total duration}}$$

Plant Use Factor: It is the ratio of kWh generated to the product of plant capacity and the number of hours for which the plant was in operation.

$$\text{Plant Use Factor} = \frac{\text{Station output in kWh}}{\text{Plant capacity} \times \text{Hours of use}}$$

Plant Utilization Factor: It is defined as the ratio of maximum demand to the plant capacity.

$$\text{Plant Utilization Factor} = \frac{\text{Maximum demand}}{\text{Plant capacity}}$$

Firm Power: It is the power intended to be always available.

Cold Reserve: It is that reserve generating capacity which is available for service but is not in operation.

Hot Reserve: It is that reserve generating capacity which is in operation but is not in service.

Spinning Reserve: It is that generating capacity which is connected to bus and is ready to take load.

Base Load and Peak Load on Power Station:

Base Load: The unvarying load which occurs almost the whole day on the station is known as base load.

Peak Load: The various peak demands of load over and above the base load of the station is known as peak load. Nuclear power stations are used as base load stations operating at high load factors of over 80%. These meet the 'block loads' which are at the at the bottom of the load curves.

Hydro – electric stations with ample storage of water are also used for base load operation. Similarly run-off river plants during river flow periods are also used to meet the base load requirements.

Hydro-electric stations with limited storage of water are used to meet the peak loads of a system.

Steam – stations are capable of operation both as peak load and base load stations depending upon the co-ordination required with other type of stations in the system. These can operate at load factor's varying from 40 to 80%.

Example 6.1

A generating station has a connected load of 43MW and a maximum demand of 20MW; the units generated being 61.5×10^6 per annum. Calculate

(i) the demand factor and (ii) load factor.

$$\begin{aligned} \text{Sol: (i) Demand factor} &= \frac{\text{Maximum demand}}{\text{Connected load}} \\ &= \frac{20}{43} = 0.465 \end{aligned}$$

$$\begin{aligned} \text{Average demand} &= \frac{\text{Units generated / annum}}{\text{Hour in a year}} \\ &= \frac{61.5 \times 10^6}{8760} = 7020 \text{ kW} \end{aligned}$$

$$\begin{aligned} \text{(ii) Load factor} &= \frac{\text{Average demand}}{\text{Maximum demand}} \\ &= \frac{7020}{20 \times 10^3} = 0.351 = 35.1\% \end{aligned}$$



Example 6.2

The maximum demand of a hydro station is 200 MW, the annual load factor being 60%. Calculate the total electrical energy generated per year.

Sol: M.D = 200 MW = 200×10^3 kW

L.F = 0.6

Annual load factor =

$$\frac{\text{no. of units generated in a year}}{\text{MD} \times \text{no. of hrs. in a year}}$$

$\therefore$ No of units generated / year

$$= 0.6 \times 200 \times 10^3 \times 8760$$

$$= 1.2 \times 876 \times 10^6$$

$$= 1051.2 \times 10^6 \text{ kWh}$$

Example 6.3

A generating station supplies the following loads to various consumers:

Industrial consumer = 1500 kW ;

Commercial establishment = 750 kW

Domestic power = 100 kW ;

Domestic light = 450 kW

If the maximum demand on the station is 2500kW and the number of kWh generated per year is 45×10^5 .

Determine

(i) the diversity factor and

(ii) annual load factor.

Sol: (i) Diversity factor = $\frac{1500 + 750 + 100 + 450}{2500}$

$$= 1.12$$

Average demand = $\frac{\text{kWh generated / annum}}{\text{hours in a year}}$

$$= \frac{45 \times 10^5}{8760} = 513.7 \text{ kW}$$

(ii) Load factor = $\frac{\text{Average load}}{\text{Max. demand}}$

$$= \frac{513.7}{2500} = 0.205 = 20.5\%$$

Example 6.4

A Generating station has a maximum demand of 15000kW. The annual load factor is 50% and capacity factor is 40%. Determine the reserve capacity of the plant.

Sol: Energy generated

$$= \text{Maximum demand} \times \text{L.F} \times \text{Hours in a year}$$

$$= (15000) \times (0.5) \times (8760) \text{ kWh}$$

$$= 65.7 \times 10^6 \text{ kWh}$$

Capacity factor

$$= \frac{\text{Units generated / annum}}{\text{Plant capacity} \times \text{Hours in a year}}$$

$$\therefore \text{Plant capacity} = \frac{65.7 \times 10^6}{0.4 \times 8760} = 18750 \text{ kW}$$

Reserve capacity = Plant capacity – maximum demand

$$= 18750 - 15000$$

$$= 3750 \text{ kW}$$

6.1 Economic Load Dispatch

The economic load dispatch involves the solution of two different problems. These are unit commitment problem and economic dispatch problem.

Unit commitment problem:

Select optimally out of the available generating sources to meet the expected load and provide a specified margin of operating reserve over a specified period of time.

Economic dispatch problem: It is required to distribute the load among the generating units actually parallel with the system in such manner as to minimize the total fuel cost minute-to-minute requirements of the system.

The economic load dispatch problem applicable for fuel based units rather than hydro electric stations.



Fuel cost of a thermal power plant

- It gives the relation between fuel cost and power output of the plant
- It is an empirical formulae
- The fuel cost of an i^{th} plant is expressed as

$$C_i(P_i) = \frac{1}{2}a_i P_i^2 + b_i P_i + c_i \text{ Rs / hr}$$

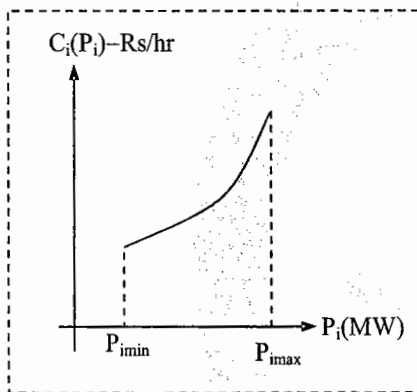
Where,

a_i, b_i , & c_i are loss coefficients of i^{th} unit

P_i = power output of i^{th} plant in Rs/hr

Fuel cost curve:

- It is a curve between fuel cost in Rs/hr versus power output unit in 'MW'



- $P_{\min} \leq P_i \leq P_{\max}$
- $P_{\min}$ is fixed by boiler and thermodynamic
- $P_{\max}$ is fixed by thermal limit of the generator

Incremental fuel cost of the unit

The incremental fuel cost is obtained by determining the increased cost of fuel for a definite time interval during which power output is increased by a small amount

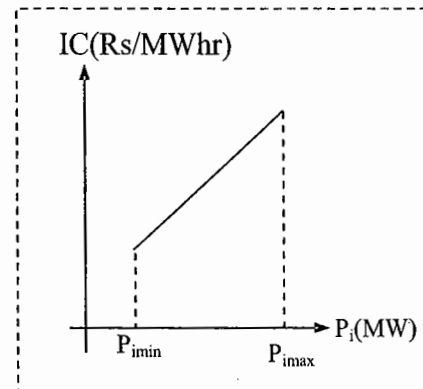
Mathematically,

$$IC_i = \frac{\partial C_i(P_i)}{\partial P_i} \text{ Rs / MWhr}$$

Where, IC_i is incremental cost of i^{th} unit

$$IC_i = a_i P_i + b_i \text{ Rs / MWhr}$$

Incremental cost curve



6.1.1. Economic Dispatch neglecting losses:

Objective:

To minimize the total fuel cost of thermal generating units.

Let us assume that total number of units are 'N'. The total fuel cost is given by

$$C_T = \sum_{i=1}^N C_i(P_i)$$

Such that

$$\sum_{i=1}^N P_i = P_D \text{ (Equality constraint)}$$

And

$$P_{\min} \leq P_i \leq P_{\max}$$

(Inequality constraint)

Using Lagrangian method, the augmented cost function F can be written as

$$F = C_T + \lambda \left(\sum_{i=1}^N P_i - P_D \right)$$

Where, λ is a Lagrangian multiplier

Minimization can be achieved by the condition

$$\frac{\partial F}{\partial P_i} = 0$$

By solving the above equation we get,

$$\frac{dC_i}{dP_i} = \lambda, i = 1, 2, \dots, N$$

Where,



$\frac{dC_i}{dP_i} = IC_i$ is the incremental cost of i^{th} generator.

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2} = \dots \dots \dots \frac{dC_N}{dP_N} = \lambda$$

Conclusion: To get the minimum fuel cost solution, the incremental fuel cost of all the units must be same.

Example 6.5:

The cost function of 50 MW generator is given by $F(P_i) = 0.02 P_i^2 + 53 P_i + 225$ Rs/hr When 100% load is applied, Calculate incremental fuel cost and fuel cost.

Sol: $F(P_i) = 0.02 P_i^2 + 53 P_i + 225$

$$\frac{dF}{dP_i} = 0.04 P_i + 53$$

$$= 0.04(50) + 53 = 55 \text{ Rs/MW hr}$$

$$\text{Fuel cost} = 0.02(50)^2 + 53(50) + 225$$

$$= 2925 \text{ Rs/hr}$$

Example 6.6:

The fuel cost of 2 units are as follows.

$$\left. \begin{aligned} F_1 &= 0.2 P_1^2 + 30 P_1 + 60 \\ F_2 &= 0.15 P_2^2 + 20 P_2 + 80 \end{aligned} \right\} \text{Rs/hr}$$

Calculate the generation schedule of the each unit for a Lagrangian multiplier of Rs.120 per MW hr

Sol: $F_1 = 0.2 P_1^2 + 30 P_1 + 60$

$$F_2 = 0.15 P_2^2 + 20 P_2 + 80$$

$$\lambda = 120 \text{ Rs/MW hr}$$

$$\frac{dF_1}{dP_1} = 0.4 P_1 + 30$$

$$\frac{dF_2}{dP_2} = 0.3 P_2 + 20$$

From co-ordination equation: $IC_1 = IC_2 = \lambda$

$$IC_1 = \lambda \Rightarrow 0.4 P_1 + 30 = 120$$

$$P_1 = \frac{120 - 30}{0.4} = \frac{90}{0.4} = 225 \text{ MW}$$

$$IC_2 = \lambda \Rightarrow 0.3 P_2 + 20 = 120$$

$$P_2 = \frac{120 - 20}{0.3} = 333.33 \text{ MW}$$

$$P_D = P_1 + P_2 = 225 + 333.33 = 558.33 \text{ MW}$$

Example 6.7:

The incremental cost characteristics of the two units in a plant are given by

$$IC_1 = 0.1 P_1 + 8.0 \text{ Rs/MW hr}$$

$$IC_2 = 0.15 P_2 + 3.0 \text{ Rs/MW hr}$$

The optimum sharing of load when the total load is 100 MW is

Sol: $IC_1 = 0.1 P_1 + 8.0 \text{ Rs/MW hr}$

$$IC_2 = 0.15 P_2 + 3.0 \text{ Rs/MW hr}$$

$$P_D = 100 \text{ MW}$$

From co ordination equation: $IC_1 = IC_2$

$$0.1 P_1 - 0.15 P_2 = -5 \text{ ----- (1)}$$

From equality constraints: $P_D = P_g = P_1 + P_2$

$$P_1 + P_2 = 100 \text{ ----- (2)}$$

From eq (1) and eq (2)

$$P_1 = 40 \text{ MW}, \quad P_2 = 60 \text{ MW}$$

Example 6.8:

The Incremental fuel cost of three plants are given as

$$IC_1 = 0.12 P_1 + 16 \text{ Rs/MW hr},$$

$$IC_2 = 0.2 P_2 + 24 \text{ Rs/MW hr},$$

$$IC_3 = 0.15 P_3 + 18 \text{ Rs/MW hr}$$

Determine the generation schedule of the 3 plants for the cost received for each plant is Rs. 30/MW hr.

Sol: From co ordination equation:

$$IC_1 = IC_2 = IC_3 = \lambda$$

$$IC_1 = \lambda \Rightarrow 0.12 P_1 + 16 = 30$$

$$P_1 = 116.66 \text{ MW}$$

$$IC_2 = \lambda \Rightarrow 0.2 P_2 + 24 = 30$$

$$P_2 = 30 \text{ MW}$$

$$IC_3 = \lambda \Rightarrow 0.15 P_3 + 18 = 30$$

$$P_3 = 80 \text{ MW}$$



Example 6.9

Incremental fuel costs in Rs/MWhr for a plant consisting of two units are

$$\frac{dC_1}{dP_1} = 0.2P_1 + 40$$

$$\frac{dC_2}{dP_2} = 0.3P_2 + 30.$$

Assuming maximum and minimum loads on each unit to be 125 MW and 20 MW respectively, the optimal load sharing by G_1 while delivering 50 MW load is

Sol: Given

$$IC_1 = 0.2 P_1 + 40$$

$$IC_2 = 0.3 P_2 + 30$$

$$20 \leq P_1 \leq 125$$

From co ordination equation

$$IC_1 = IC_2 \Rightarrow 0.2 P_1 - 0.3 P_2 = -10 \text{ ----- (i)}$$

From equality constant

$$P_1 + P_2 = 50 \text{ W} \text{ ----- (ii)}$$

Solving i & ii

$$P_1 = 10 \text{ MW} ; P_2 = 40 \text{ MW}$$

$$P_1 < P_{1\min} \Rightarrow P_1 < 20 \text{ MW}$$

$$\text{So } P_1 = 20 \text{ MW} ; P_2 = 30 \text{ MW}$$

Example 6.10:

The Fuel inputs of plant 1 & 2 are given as

$$F_1 = 0.2 P_1^2 + 40 P_1 + 120 \text{ Rs/hr}$$

$$F_2 = 0.25 P_2^2 + 30 P_2 + 150 \text{ Rs/hr}$$

The minimum & max load on each plant is

$$25 \leq P_i \leq 100 \text{ Where } i = 1, 2.$$

Determine the economical operating schedule and corresponding cost of generation for a total demand of 180 MW.

Sol: $IC_1 = 0.4 P_1 + 40 \text{ Rs/MWhr}$

$$IC_2 = 0.5 P_2 + 30 \text{ Rs/MWhr}$$

$$25 \leq P_1 \text{ \& } P_2 \leq 120 \text{ MW} ;$$

$$P_D = 180 \text{ MW}$$

$$IC_1 = IC_2 \Rightarrow 0.4 P_1 - 0.5 P_2 = 10 \text{ ----- (i)}$$

$$P_1 + P_2 = 180 \text{ ----- (ii)}$$

Solving i & ii

$$P_1 = 111.11 \text{ MW}$$

$$P_2 = 68.89 \text{ MW}$$

$$F_T = F_1 (111.11) + F_2 (68.89)$$

$$= 0.2 (111.11)^2 + 40 (111.11) + 120 +$$

$$0.25 (68.89)^2 + 30 (68.89) + 150$$

$$= 10,436.64 \text{ Rs/hr}$$

6.1.2. Economic Dispatch with power loss:

Objective:

To minimize the total fuel cost of thermal generating units.

Let us assume that total number of units are 'N'. The total fuel cost is given by

$$C_T = \sum_{i=1}^N C_i(P_i)$$

Such that,

$$\sum_{i=1}^N P_i = P_D + P_L \text{ (Equality constraint)}$$

Where,

P_L is the power loss in 'MW' and it is expressed as

$$P_L = \sum_{i=1}^N \sum_{j=1}^N P_i B_{ij} P_j \text{ MW}$$

For $N = 2$, the power loss formulae expressed as

$$P_L = B_{11} P_1^2 + B_{22} P_2^2 + 2 B_{12} P_1 P_2 \text{ MW}$$

And

$P_{\min} \leq P_i \leq P_{\max}$ (Inequality constraint) Using *Lagrangian method*, the augmented cost function F can be written as

$$F = C_T + \lambda \left(\sum_{i=1}^N P_i - P_D - P_L \right)$$

Where, λ is a Lagrangian multiplier

Minimization can be achieved by the condition

$$\frac{\partial F}{\partial P_i} = 0$$

By solving the above equation,

$$L_i \frac{dC_i}{dP_i} = IC_i$$



Where, $L_i = \frac{1}{1 - \frac{\partial P_L}{\partial P_i}}$,

$\frac{\partial P_L}{\partial P_i}$ is the incremental transmission loss the i^{th} unit and it is expressed as,

$$\frac{\partial P_L}{\partial P_i} = \sum_{j=1}^N 2(B_{ij} P_j)$$

$$L_1 \frac{dC_1}{dP_1} = L_2 \frac{dC_2}{dP_2} = \dots \dots \dots L_N \frac{dC_N}{dP_N} = \lambda$$

6.1.3 Special cases

Case : 1 if the load is located at plant – 1



$$R_1 = 0 \quad R_2 \neq 0 \quad R_3 = 0$$

$$B_{11} \propto (R_1 + R_3) \Rightarrow B_{11} = 0$$

$$B_{22} \propto (R_2 + R_3) \Rightarrow B_{22} \neq 0$$

$$(B_{12} = B_{21}) \propto R_3 \Rightarrow B_{12} = 0$$

$$P_L = B_{11} P_1^2 + B_{22} P_2^2 + 2 B_{12} P_1 P_2 \text{ MW}$$

$$P_L = B_{22} P_2^2$$

$$\frac{\partial P_L}{\partial P_1} = 0; \quad \frac{\partial P_L}{\partial P_2} = 2 B_{22} P_2$$

$$L_1 = \frac{1}{1 - \frac{\partial P_L}{\partial P_1}} = \frac{1}{1 - 0} = 1$$

$$L_2 = \frac{1}{1 - \frac{\partial P_L}{\partial P_2}} = \frac{1}{1 - 2 B_{22} P_2} > 1$$

Case 2: If load is located at plant -2



$$R_1 \neq 0; \quad R_2 = 0; \quad R_3 = 0$$

$$B_{11} \propto (R_1 + R_3) \Rightarrow B_{11} \neq 0$$

$$B_{22} \propto (R_2 + R_3) \Rightarrow B_{22} = 0$$

$$(B_{12} = B_{21}) \propto R_3 \Rightarrow B_{12} = 0$$

$$P_L = B_{11} P_1^2$$

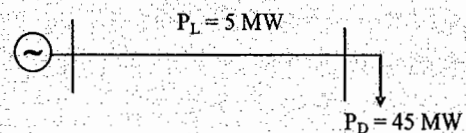
$$\frac{\partial P_L}{\partial P_1} = 2 B_{11} P_1; \quad \frac{\partial P_L}{\partial P_2} = 0$$

$$L_1 \frac{1}{1 - \frac{\partial P_L}{\partial P_1}} = \frac{1}{1 - 2 B_{11} P_1} > 1$$

$$L_2 = \frac{1}{1 - \frac{\partial P_L}{\partial P_2}} = \frac{1}{1 - 0} = 1$$

Example 6.11:

The penalty factor of the system shown below is



Sol: $P_g = P_D + P_L = 45 + 5 = 50 \text{ MW}$

$$I.T.L = \frac{P_L}{P} = \frac{5}{50} = 0.1$$

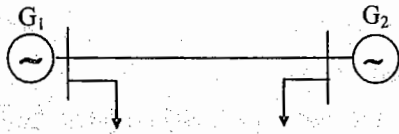
Penalty Factor,

$$L = \frac{1}{1 - I.T.L} = \frac{1}{1 - 0.1} = 1.11$$



Example 6.12:

A two bus power system is shown below. The incremental fuel cost of the two plants are



$$IC_1 = 0.35 P_1 + 41 \text{ Rs/MWhr}$$

$$IC_2 = 0.35 P_2 + 41 \text{ Rs/MWhr}$$

The loss expression is $P_L = 0.001 (P_2 - 70)^2 \text{ MW}$

The total incremental cost of the system is 117.6 Rs/MWhr

(i) Calculate the optimal scheduling of generators

(ii) Calculate the total power loss.

Sol: (i) From co-ordination equation

$$\lambda = L_1 IC_1 = L_2 IC_2 \text{ -----(i)}$$

$$P_L = 0.001 (P_2 - 70)^2 \text{ MW}$$

$$\frac{\partial P_L}{\partial P_1} = 0; \frac{\partial P_L}{\partial P_2} = 0.002 P_2 - 0.14$$

$$L_1 = \frac{1}{1 - \frac{\partial P_L}{\partial P_1}} = 1;$$

$$L_2 = \frac{1}{1 - \frac{\partial P_L}{\partial P_2}} = \frac{1}{1 - (0.002 P_2 - 0.14)}$$

$$= \frac{1}{1.14 - 0.002 P_2}$$

$$L_1 IC_1 = \lambda$$

$$(0.35 P_1 + 41) = 117.6$$

$$P_1 = 218.85 \text{ MW}$$

$$L_2 IC_2 = \lambda$$

$$\left(\frac{1}{1.14 - 0.002 P_2} \right) (0.35 P_2 + 41) = 117.6$$

$$P_2 = 159.029 \text{ MW}$$

$$(ii) P_L = 0.001 (P_2 - 70)^2$$

$$= 0.001 (159.029 - 70)^2$$

$$= 7.92 \text{ MW}$$

$$P_D = P_g - P_L = 369.966 \text{ MW}$$